

Course: B. Tech. Branch: Computer Engineering/CSE Semester: VII

Subject Code & Name: Blockchain Technology (BTCOE704A)

Max Marks: 60

Date:20/07/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What are Cryptographic Hash Functions? And explain the properties of the hash function.	Understand	6
B) What are the key properties and features of smart contracts?	Understand	6
C) Define Blockchain technology and explain the types of Blockchain Technology.	Understand	6
Q.2 Solve Any Two of the following.		12
A) Write the difference between Proof of Work (PoW) and Proof of Stake (PoS) in blockchain.	Understand	6
B) How does a Bitcoin transaction progress from creation to confirmation in the Bitcoin network?	Analysis	6
C) What is Double Spending in Blockchain? Explain with an example.	Apply	6
Q. 3 Solve Any Two of the following.		12
A) What challenges arise in designing a permissioned blockchain?	Apply	6
B) What are the key safety rules in the Raft consensus protocol that ensure consistency and integrity in the replicated log of nodes?	Understand	6
C) What are the differences between permissioned and permissionless blockchains?	Analysis	6
Q.4 Solve Any Two of the following.		12
A) What potential benefits can the mortgage industry gain from adopting blockchain technology?	Apply	6
B) Give any examples of uses for blockchain technology, along with a justification.	Apply	6
C) What is KYC? Explain the Shared KYC model based on Blockchain.	Understand	6
Q. 5 Solve Any Two of the following.		12

A)	What are the key components of the Hyperledger Fabric architecture, and how do they contribute to the functionality and security of enterprise-grade blockchain networks?	Understand	6
B)	What is the difference between Ripple and Corda?	Analysis	6
C)	What are the key differences between Hyperledger Fabric and Ethereum?	Analysis	6

*** All the Best ***